INCH-POUND

MIL-PRF-49465/7B
11 December 2000
SUPERSEDING
MIL-PRF-49465/7A
6 January 1992

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, METAL ELEMENT (POWER TYPE), STYLE RLV31

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for style RLV31 metal element, power type, fixed resistors.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of a basic style of this specification and a coded number. The PIN is in the following form:

M4946507	TR1000F
Associated Specification	Coded number (see MIL-PRF-49465)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, Ohio 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-49465 - Resistors, Fixed, Metal Element (Power Type), (Very Low Resistance Values), General Specification for.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the test of this document and the reference cited herein, the test of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-49465.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1.

3.3 Characteristic. This style of resistor is available in characteristic T as described in MIL-PRF-49465.

3.4 Tolerance. This style of resistor is available in tolerances F, H, and J (±1 percent, ±3 percent, and ±5 percent, respectively).

3.5 Terminal lead types. This style of resistor is a two-leaded device with solderable terminals (see figure 1).

3.6 Weight. The weight shall be .01 lbs. (5.0 grams) as shown in figure 1.

3.7 Power rating. The power rating shall be 5.0 watts at +25°C as shown in table I.

TABLE I. Electrical characteristics.

Style	Rated wattage at 25°C (watts)	Minimum resistance (ohms)	Maximum resistance (ohms)	Maximum overload current (amperes)
RLV31	5.0	0.01	0.30	40.0

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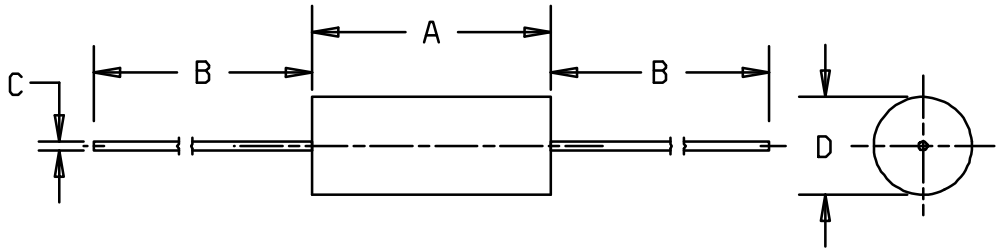
3.8 Maximum overload current. The maximum overload current shall be as specified in table I.

3.9 Resistance value range. The minimum and maximum resistance values shall be as specified in table I.



Style	Maximum weight lbs. (grams)	Dimensions			
		A	B	C	D
RLV31	.01 (5.00)	.925 ±.031 (23.50 ±.787)	1.50 min (38.10 min)	.040 ±.005 (1.02 ±.130)	.330 ±.031 (8.38 ±.787)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. For resistance measurements the point of contact to the leads shall be .375 ±.062 (9.52 mm ±1.57 mm) from the body.
5. The picturization of the style above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.

FIGURE 1. Resistors, fixed, metal element.

3.10 Temperature coefficient. The temperature coefficient of resistance shall be as specified in table II.

3.11 Marking. The resistors shall be marked in accordance with MIL-PRF-49465.

TABLE II. Temperature coefficient of resistance.

Style	.01 ohm to .0249 ohm	.025 ohm to .0499 ohm	.05 ohm to .0749 ohm	.075 ohm to .099 ohm	0.1 ohm and above
RLV31	±250 ppm/°C	±150 ppm/°C	±100 ppm/°C	±75 ppm/°C	±50 ppm/°C

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-49465.

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4.2 Visual and mechanical inspection. Visual and mechanical inspection shall be in accordance with MIL-PRF-49465.

4.3 Terminal strength. Terminal strength shall be in accordance with MIL-PRF-49465 and the applied force shall be five pounds.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-49465 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification, the applicable associated specification, and complete PIN (see 1.2.1 and 3.1).

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

d. Bracket assembly requirements (see 6.5 of MIL-PRF-49465).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
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Preparing activity:
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE
	MIL-PRF-49465/7B	11 December 2000

3. DOCUMENT TITLE

Resistors, Fixed, Metal Element (Power Type), Style RLV31.

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Andrew Ernst	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0552 850-0552 614-692-6939 andrew_ernst@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC -LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvior, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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